# Section 15.3: Double Integrals in Polar Coordinates

Multiple Integrals

MTH 310
Calculus III

## Why Polar?

Not every region fits nicely into Cartesian coordinates. Consider integrating over a disk like $x^2 + y^2 \leq 4$.

In Cartesian, the bounds are:

$$\int_{-2}^{2} \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} f(x, y)\, dy\, dx$$

That square root makes everything painful — and the region isn't $x$-simple or $y$-simple in a clean way.

But in **polar coordinates**, a disk of radius 2 is just $0 \leq r \leq 2$, $0 \leq \theta \leq 2\pi$. The bounds are **constants**!

Regions described by circles, sectors, and cardioids are natural in polar. When the region has circular symmetry, polar coordinates will simplify the integral dramatically.

## Polar Rectangles and the Area Element

In Cartesian coordinates, we chop the plane into small rectangles of area $\Delta A = \Delta x\, \Delta y$.

In polar coordinates, we chop the plane into **polar rectangles**: regions bounded by two radii and two arcs.

A polar coordinate grid showing concentric circles at different radii and radial lines at different angles, forming polar rectangles. One polar rectangle is highlighted and labeled, showing it is bounded by r = r1 and r = r2 on the sides and theta = theta1 and theta = theta2 on the arcs. The highlighted rectangle is wider at larger r than at smaller r.

A polar rectangle with $r$ from $r_i$ to $r_i + \Delta r$ and $\theta$ from $\theta_j$ to $\theta_j + \Delta\theta$ has area:

$$\Delta A_i \approx r_i\, \Delta r\, \Delta\theta$$

The key insight: polar rectangles are **wider** at larger $r$. The factor of $r$ accounts for this stretching. A small $\Delta r\, \Delta\theta$ box near $r = 10$ covers more area than the same box near $r = 1$.

In polar coordinates, $dA = r\, dr\, d\theta$ — **not** just $dr\, d\theta$. The extra $r$ is essential.

## The Polar Double Integral

**Double Integral in Polar Coordinates**

If $f$ is continuous on a polar region $R$ described by $\alpha \leq \theta \leq \beta$ and $h_1(\theta) \leq r \leq h_2(\theta)$, then:

$$\iint_R f(x, y)\, dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r\cos\theta,\ r\sin\theta)\ r\, dr\, d\theta$$

**Three things change** when converting to polar:

1. Replace $x$ with $r\cos\theta$ and $y$ with $r\sin\theta$
2. Replace $dA$ with $r\, dr\, d\theta$ (don't forget the $r$!)
3. Replace the Cartesian bounds with polar bounds

The most common mistake is forgetting the extra $r$ in $dA = r\, dr\, d\theta$. Write it down every time.

## Quick Check: Convert to Polar

Set up (but **do not evaluate**) the polar integral for:

$$\iint_R (x^2 + y^2)\, dA$$

where $R$ is the unit disk $x^2 + y^2 \leq 1$.

*What replaces $x^2 + y^2$? What replaces $dA$? What are the bounds?*

*Use $x^2 + y^2 = r^2$ and $dA = r\,dr\,d\theta$. The unit disk is $0 \leq r \leq 1$, $0 \leq \theta \leq 2\pi$.*

## Example 1: Area of $r = 4\cos\theta$ in QI

**Example 1**

Find the area in Quadrant I enclosed by the graph of $r = 4\cos\theta$.

First, what does $r = 4\cos\theta$ look like? Multiply both sides by $r$: $r^2 = 4r\cos\theta$, so $x^2 + y^2 = 4x$, which gives $(x-2)^2 + y^2 = 4$. It's a **circle of radius 2 centered at $(2, 0)$**.

*Area = $\iint_R 1\,dA$ in polar. The curve starts at $\theta = 0$ (where $r = 4$) and returns to the origin at $\theta = \pi/2$ (where $r = 0$). Inner bounds: $0 \leq r \leq 4\cos\theta$.*

## Example 2: Volume of a Hemisphere

**Example 2**

Evaluate $\iint_R \sqrt{4 - x^2 - y^2}\, dA$ where $R = \{(x, y) \mid x^2 + y^2 \leq 4,\ x \geq 0\}$.

The integrand $z = \sqrt{4 - x^2 - y^2}$ is the **top half of the sphere** $x^2 + y^2 + z^2 = 4$ (radius 2). The region $R$ is a **semicircle** (right half of the disk of radius 2).

*Convert to polar: $\sqrt{4 - x^2 - y^2} = \sqrt{4 - r^2}$. The semicircle where $x \geq 0$ has $-\pi/2 \leq \theta \leq \pi/2$ and $0 \leq r \leq 2$.*

## Quick Check: Sketch a Polar Region

*Sketch the region described by $1 \leq r \leq 2$ and $0 \leq \theta \leq \pi/4$. What shape is it?*

*Mark the radii $r = 1$ and $r = 2$ and the angles $\theta = 0$ and $\theta = \pi/4$. Shade the region between them.*

## Example 3: Area of the Lima\c{c}on $r = 4 + 3\cos\theta$

**Example 3**

Find the area enclosed by $r = 4 + 3\cos\theta$.

Since $4 > 3$, this is a lima\c{c}on **without an inner loop** — it's a slightly flattened, egg-shaped curve. The curve is traced once as $\theta$ goes from $0$ to $2\pi$.

*Area = $\int_0^{2\pi} \int_0^{4+3\cos\theta} r\,dr\,d\theta$. Evaluate the inner integral first, then use the identity $\cos^2\theta = \frac{1+\cos 2\theta}{2}$.*

## Example 4: Volume Inside a Sphere and Outside a Cylinder

**Example 4**

Find the volume of the solid inside the sphere $x^2 + y^2 + z^2 = 16$ and outside the cylinder $x^2 + y^2 = 4$.

**Visualize:** The sphere has radius 4. The cylinder has radius 2. We want the volume of the sphere with a cylindrical hole of radius 2 drilled through the center.

**Set up:** By symmetry, compute the volume above the $xy$-plane and double it.

Above the $xy$-plane, $z = \sqrt{16 - x^2 - y^2} = \sqrt{16 - r^2}$.

The region in the $xy$-plane: outside the cylinder means $r \geq 2$; inside the sphere means $r \leq 4$.

*Use $V = 2\int_0^{2\pi} \int_2^4 \sqrt{16 - r^2}\, r\,dr\,d\theta$. The inner integral uses $u = 16 - r^2$.*

## When to Use Polar Coordinates

How do you know when to switch to polar? Here are the telltale signs:

| Clue | Example |
| --- | --- |
| Region is a disk, annulus, or sector | $x^2 + y^2 \leq 9$ |
| Boundary involves $x^2 + y^2$ | $1 \leq x^2 + y^2 \leq 4$ |
| Integrand contains $x^2 + y^2$ | $e^{-(x^2+y^2)}$ |
| Boundary is a polar curve | $r = 1 + \cos\theta$ |
| Square-root bounds like $\sqrt{a^2 - x^2}$ | Semicircle or quarter-circle region |

**Conversion checklist:**

1. Replace $x = r\cos\theta$, $y = r\sin\theta$, $x^2 + y^2 = r^2$
2. Replace $dA = dy\,dx$ with $r\,dr\,d\theta$ (**don't forget the $r$!**)
3. Determine polar bounds for $r$ and $\theta$
4. Evaluate (inner $r$ first, then outer $\theta$)

Not every integral is easier in polar. If the region is a rectangle or the integrand involves $x$ and $y$ separately (not in $x^2 + y^2$ combinations), Cartesian may be simpler.

## Homework

**Section 15.3:** 1-7, 9, 11, 13, 17, 19, 21, 23, 25, 27, 29, 31, 50a